Rafael Antonio Tanda Pretel

Software Engineer — Backend, offline-first systems & AI-assisted delivery

Havana, Cuba · relocating to La Palma, Spain · Spanish citizen · EU work authorization
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PROFILE

Software engineer who ships complete products for real users, from requirements to tests, packaging and support. Backends in Go and TypeScript; offline-first clients in Flutter, Godot and Python for low-end hardware and unreliable networks. I work AI-first: I direct coding agents through written specs, reviewable units and tests and own every line that ships.

FREELANCE EXPERIENCE

Salon Manager — offline-first mobile app · Freelance · hair salon — Sep 2026

`Flutter, Dart, sqflite, Provider · github.com/Ranta2025/gestion-salon`

- Gathered requirements with the owner and delivered finances, clients, appointments with reminders, cash closing and PDF/CSV reports, fully on-device with no backend or recurring cost.
- Repositories tested against real in-memory SQLite; restore validates the backup before touching data.

Salary Fund Distribution — Acuerdo 10199/2025 · Freelance · client organization — Oct 2026

`Python, tkinter, pytest, PyInstaller · github.com/Ranta2025/fondo-salario-acuerdo-10199`

- Turned a 2025 Cuban regulation into an offline desktop app that distributes unexecuted salary funds, with configurable weights so each unit can match its own rules.
- Totals reconcile to the cent (largest-remainder rounding); domain logic isolated and covered by 100+ pytest tests; auditable Excel export.

AI-ASSISTED ENGINEERING

- Agent-driven workflow: requirements become numbered specs; coding agents (Claude Code, opencode) implement them in reviewable units; every change passes tests, multi-lens AI review and my own diff review.
- Built agent-bridge (github.com/Ranta2025/agent-bridge, MIT): links Claude Code to gentle-shell agents (orchestrated by gentle-ai) and tracks tasks to completion — state machine with delivery confirmation and retries, policy-driven auto-approver that never allows push, force or secret paths, RAM guard, 197 tests.
- Extended the tools: a Claude Code side-panel mod, an opencode background-agents plugin, and custom skills such as one that makes the agent challenge a design with evidence before building.

SELECTED PROJECTS

ChatApp — real-time messaging platform — Jan 2026 – present

`Go, Gin, React 19, PostgreSQL, Redis, MinIO, WebSockets, Docker, Playwright · github.com/Ranta2025/chatapp`

- Designed a 7-service platform: 1:1 and group chat, receipts, presence, media, voice notes, video calls and Web Push; group permissions enforced server-side.
- Hardened auth (15-min JWTs in HttpOnly cookies, rotated refresh tokens, lockout, rate limits); PWA offline outbox delivers queued messages in order; 276 test files plus Playwright e2e in CI.

OrtoSim CUBA — offline medical training game (undergraduate thesis) — Oct 2026 – present

`Godot 4, GDScript, GUT, PocketBase, Caddy, JSON Schema, Docker`

- Serious game for 2 GB RAM Android phones under hard budgets (< 60 MB APK, 30 FPS, < 100 draw calls); hexagonal domain behind five ports, 150 GUT test files.
- Data-driven case engine (JSON Schema, 12 step kinds) shipped as signed content packs with a SHA-256 manifest; offline event queue synced via PocketBase.

SKILLS

Languages	Go, TypeScript, Python, Dart, GDScript, Java
Backend & data	Gin, REST, WebSockets, PostgreSQL, Redis, SQLite, PocketBase, S3/MinIO
Clients	React 19, PWA (service workers, IndexedDB, Web Push), Flutter, Godot 4
AI tooling	Claude Code, opencode, MCP, agent skills/hooks/plugins, spec-driven dev, AI review
Quality & delivery	go test, Playwright, pytest, GUT, GitHub Actions, Docker Compose, Nginx, Caddy

EDUCATION · LANGUAGES

Education	B.Eng. in Computer Science, University of Information Sciences (UCI), Havana — 3rd of 4 years, expected 2028
Languages	Spanish — native · English — B1 (certification exam March 2027)